Statistics

Measures of Central Tendency

Mean: The sum of all values divided by the number of values.

$$\bar{x} = \frac{\sum x}{n} \qquad \text{or for a frequency table:} \qquad \bar{x} = \frac{\sum fx}{\sum f}$$

Median: The middle value when data is arranged in order. For n values, the median is the $\frac{n+1}{2}$th value.

Mode: The most frequently occurring value. A data set can be bimodal (two modes) or have no mode.

Exam Tip: The mean is affected by outliers. The median is resistant to outliers and often better for skewed data. Always state which measure you are using and why.

Measures of Spread

Range and Interquartile Range

Range = highest value − lowest value.

Interquartile range (IQR) = $Q_3 - Q_1$, where Q_1 is the lower quartile and Q_3 is the upper quartile.

Variance and Standard Deviation

Variance:

$$\mathrm{Var}(X) = \sigma^2 = \frac{\sum x^2}{n} - \bar{x}^2 = \frac{\sum (x - \bar{x})^2}{n}$$

Standard deviation: $\sigma = \sqrt{\mathrm{Var}(X)}$

Using summary statistics $\sum x$ and $\sum x^2$:

$$\sigma^2 = \frac{\sum x^2}{n} - \left(\frac{\sum x}{n}\right)^2$$

Worked Example

Data has $\sum x = 120$, $\sum x^2 = 1960$, $n = 10$. Find the variance.

Solution:

$$\sigma^2 = \frac{1960}{10} - \left(\frac{120}{10}\right)^2 = 196 - 144 = 52$$

Outliers

A common rule: a value is an outlier if it is more than $1.5 \times \mathrm{IQR}$ beyond the quartiles:

- Below $Q_1 - 1.5 \times \mathrm{IQR}$
- Above $Q_3 + 1.5 \times \mathrm{IQR}$

Coding

Coding simplifies calculations by transforming data. If $y = \frac{x-a}{b}$, then:

$$\bar{x} = b\bar{y} + a \qquad \sigma_x = b\sigma_y$$

Worked Example

Data is coded as $y = \frac{x-50}{10}$. If $\bar{y} = 2$ and $\sigma_y = 1.5$, find $\bar{x}$ and σ_x.

$$\bar{x} = 10(2) + 50 = 70 \qquad \sigma_x = 10(1.5) = 15$$

Exam Tip: Coding affects the mean (both shift and scale) but the standard deviation is only affected by the scale factor b, not the shift a.

Correlation and Regression

Scatter Diagrams

A scatter diagram shows the relationship between two variables. Look for:

- **Positive correlation:** as one increases, the other tends to increase
- **Negative correlation:** as one increases, the other tends to decrease
- **No correlation:** no clear linear trend

Product Moment Correlation Coefficient (PMCC)

The PMCC r measures the strength and direction of linear correlation.

$$-1 \leq r \leq 1$$

- $r = 1$: perfect positive linear correlation
- $r = -1$: perfect negative linear correlation
- $r = 0$: no linear correlation

Regression Line

The regression line of y on x is $y = a + bx$, where b is the gradient and a is the intercept.

The regression line always passes through $(\bar{x}, \bar{y})$.

Exam Tip: Only use the regression line for **interpolation** (within the range of the data). **Extrapolation** (outside the range) is unreliable. Also, correlation does not imply causation.

Probability

Basic Probability

Complement: $P(A') = 1 - P(A)$
Addition law: $P(A \cup B) = P(A) + P(B) - P(A \cap B)$
Mutually exclusive: $P(A \cap B) = 0$, so $P(A \cup B) = P(A) + P(B)$

Independent Events

Events A and B are independent if and only if:

$$P(A \cap B) = P(A) \times P(B)$$

Conditional Probability

$$P(A|B) = \frac{P(A \cap B)}{P(B)}$$

This reads: "the probability of A given that B has occurred."

Tree Diagrams and Bayes' Theorem

For two events A and A' with outcomes B:

$$P(B) = P(B|A)P(A) + P(B|A')P(A')$$

$$P(A|B) = \frac{P(B|A)P(A)}{P(B)}$$

Worked Example

$P(A) = 0.6$, $P(B|A) = 0.5$, $P(B|A') = 0.3$. Find $P(B)$ and $P(A|B)$.

$$P(B) = 0.5 \times 0.6 + 0.3 \times 0.4 = 0.30 + 0.12 = 0.42$$

$$P(A|B) = \frac{0.5 \times 0.6}{0.42} = \frac{0.30}{0.42} = \frac{5}{7} \approx 0.714$$

Discrete Probability Distributions

Binomial Distribution

If $X \sim B(n, p)$ (number of successes in n independent trials, each with probability p):

$$P(X = r) = \binom{n}{r} p^r (1-p)^{n-r}$$

$$\text{Mean} = np \qquad \text{Variance} = np(1-p)$$

Conditions for binomial: fixed number of trials, independent trials, two outcomes (success/failure), constant probability.

Worked Example

$X \sim B(10, 0.3)$. Find $P(X = 3)$.

$$P(X = 3) = \binom{10}{3}(0.3)^3(0.7)^7 = 120 \times 0.027 \times 0.0824 \approx 0.267$$

Poisson Distribution

If $X \sim \mathrm{Po}(\lambda)$ (number of events in a fixed interval, mean rate λ):

$$P(X = r) = \frac{e^{-\lambda}\lambda^r}{r!}$$

$$\text{Mean} = \lambda \qquad \text{Variance} = \lambda$$

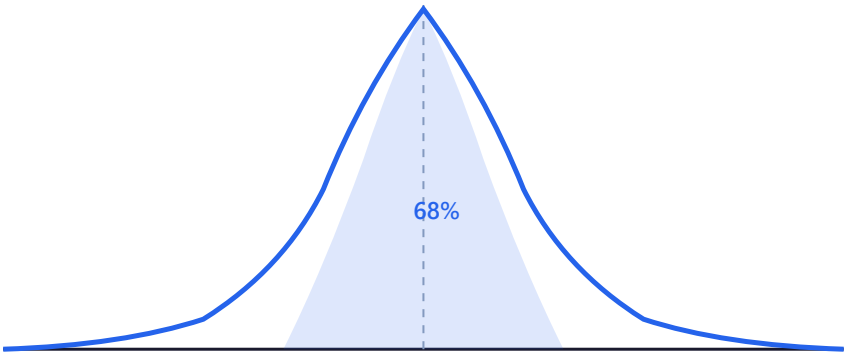
Conditions: events occur independently, at a constant average rate, and singly (not simultaneously).

Exam Tip: For Poisson, the mean equals the variance. If data shows mean $\approx$ variance, it suggests a Poisson model. A binomial can be approximated by Poisson when n is large and p is small, with $\lambda = np$.

Normal Distribution

The normal distribution $X \sim N(\mu, \sigma^2)$ is symmetric and bell-shaped about the mean μ.

- $\approx 68\%$ of data within $\mu \pm \sigma$
- $\approx 95\%$ of data within $\mu \pm 2\sigma$
- $\approx 99.7\%$ of data within $\mu \pm 3\sigma$



Standard Normal Distribution

Standardise using:

$$Z = \frac{X - \mu}{\sigma} \qquad \text{where } Z \sim N(0, 1)$$

Worked Example

$X \sim N(50, 16)$. Find $P(X < 54)$.

Here $\sigma = \sqrt{16} = 4$.

$$Z = \frac{54 - 50}{4} = 1 \implies P(X < 54) = P(Z < 1) = 0.8413$$

Finding Unknown Normal Parameters

When given a probability and asked to find μ or σ, standardise and use the inverse normal (percentage point) tables.

Worked Example: Finding σ

$X \sim N(60, \sigma^2)$ and $P(X > 70) = 0.05$. Find σ.

Solution: From tables, $P(Z > z) = 0.05 \implies z = 1.6449$.

$$\frac{70 - 60}{\sigma} = 1.6449 \implies \sigma = \frac{10}{1.6449} \approx 6.08$$

Worked Example: Finding μ

$X \sim N(\mu, 25)$ and $P(X < 47) = 0.9$. Find μ.

Solution: $\sigma = 5$. From tables, $\Phi^{-1}(0.9) = 1.2816$.

$$\frac{47 - \mu}{5} = 1.2816 \implies \mu = 47 - 6.408 = 40.6 \text{ (3 s.f.)}$$

Worked Example: Finding Both μ and σ

$X \sim N(\mu, \sigma^2)$ with $P(X < 50) = 0.8$ and $P(X < 40) = 0.3$. Find μ and σ.

Solution: Standardise both:

$$\frac{50 - \mu}{\sigma} = 0.8416 \qquad \frac{40 - \mu}{\sigma} = -0.5244$$

Subtract the equations: $\frac{10}{\sigma} = 1.3660 \implies \sigma = 7.32$.

Then $\mu = 50 - 0.8416 \times 7.32 = 43.8$.

Exam Tip: When $P(X < x) < 0.5$, the z-value is negative. For $P(X > x)$, note $P(Z > z) = P(Z < -z)$ by symmetry. Always sketch the curve and shade before reading from tables.

Distribution Approximations

From	To	Condition	Parameters
$B(n, p)$	$\mathrm{Po}(\lambda)$	n large, p small	$\lambda = np$
$B(n, p)$	$N(\mu, \sigma^2)$	n large, $np > 5$, $nq > 5$	$\mu = np$, $\sigma^2 = np(1-p)$
$\mathrm{Po}(\lambda)$	$N(\mu, \sigma^2)$	λ large ($\lambda > 15$)	$\mu = \lambda$, $\sigma^2 = \lambda$

Exam Tip: When approximating a discrete distribution with a normal, apply a **continuity correction**: for $P(X \leq k)$, calculate $P(Y \leq k + 0.5)$. For $P(X \geq k)$, calculate $P(Y \geq k - 0.5)$.

Hypothesis Testing

Null hypothesis H_0: the default assumption (e.g. $p = 0.3$, $\mu = 50$).
Alternative hypothesis H_1: what we are testing for (e.g. $p > 0.3$, $\mu \neq 50$).

Steps for a Hypothesis Test

1. State H_0 and H_1
2. Choose the significance level (e.g. 5%)
3. Calculate the test statistic
4. Find the critical value or p-value
5. Compare and decide: reject or do not reject H_0
6. Write a conclusion in context

One-tailed vs Two-tailed Tests

- **One-tailed:** $H_1 : p > k$ or $H_1 : p < k$ — entire significance level in one tail
- **Two-tailed:** $H_1 : p \neq k$ — significance level split between both tails (e.g. 2.5% each for 5%)

Test for the Mean

For a sample mean $\bar{x}$ from a normal distribution with known variance σ^2:

$$Z = \frac{\bar{x} - \mu}{\sigma / \sqrt{n}}$$

Worked Example

A company claims mean weight is 500g. A sample of 25 gives $\bar{x} = 495$ and $\sigma = 10$. Test at 5% significance.

$H_0 : \mu = 500$, $H_1 : \mu < 500$ (one-tailed)

$$Z = \frac{495 - 500}{10/\sqrt{25}} = \frac{-5}{2} = -2.5$$

Critical value at 5% one-tailed: $Z = -1.645$. Since $-2.5 < -1.645$, reject H_0. There is significant evidence that the mean weight is less than 500g.

Type I and Type II Errors

	H_0 true	H_0 false
Reject H_0	Type I error	Correct
Do not reject H_0	Correct	Type II error

$P(\text{Type I error}) = \text{significance level}$

Binomial Hypothesis Test

Used to test a claim about a proportion p using the binomial model. Let $X \sim B(n, p)$ under H_0.

Worked Example: Binomial Hypothesis Test

A coin is claimed to be fair ($p = 0.5$). In 20 tosses, 14 heads are obtained. Test at 5% significance whether the coin is biased towards heads.

$H_0 : p = 0.5$, $H_1 : p > 0.5$ (one-tailed)

Under H_0: $X \sim B(20, 0.5)$. Find $P(X \geq 14)$:

$$P(X \geq 14) = 1 - P(X \leq 13) = 1 - 0.9423 = 0.0577$$

Since $0.0577 > 0.05$, we do not reject H_0. There is insufficient evidence at the 5% level to conclude the coin is biased towards heads.

Worked Example: Critical Region

Find the critical region for a one-tailed test ($H_1 : p > 0.3$) with $n = 20$ at the 5% level.

Find the smallest c such that $P(X \geq c) \leq 0.05$ where $X \sim B(20, 0.3)$.

From tables: $P(X \geq 10) = 0.0480 < 0.05$ and $P(X \geq 9) = 0.1133 > 0.05$.

So the critical region is $X \geq 10$. The actual significance level is 4.80%.

Exam Tip: When finding the critical region, the actual significance level will usually be less than the stated level (because the binomial is discrete). Always state both the critical region AND the actual significance level.

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Correlation Hypothesis Test

To test whether a population has linear correlation:

$H_0 : \rho = 0$ (no correlation), $H_1 : \rho \neq 0$ (or one-tailed). Compare the sample r with the critical value from tables.

Chi-Squared Goodness-of-Fit Test

The test statistic:

$$\chi^2 = \sum \frac{(O - E)^2}{E}$$

where O = observed frequency and E = expected frequency. Degrees of freedom = number of categories − 1 − number of estimated parameters.

Exam Tip: In hypothesis testing, always state your hypotheses clearly, show your working, compare with the critical value, and write a conclusion **in context** (e.g. "there is sufficient evidence at the 5% level to suggest...").

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